K-Nearest Neighbors (KNN)

What is KNN?

K-Nearest Neighbors (KNN) is a **supervised machine learning algorithm** used for **classification** and **regression**.

It makes a prediction by looking at the **nearest data points** around a new input.

Simple idea:

Similar data points are usually close to each other.

So if a new point is near points from one class, KNN predicts that it belongs to that class.

Why is it called "K-Nearest Neighbors"?

- **K** = the number of nearest points we check
- **Nearest Neighbors** = the closest data points to the new point

Example:
If **k = 3**, the algorithm looks at the **3 closest points** and decides based on them.

Type of algorithm

KNN is:

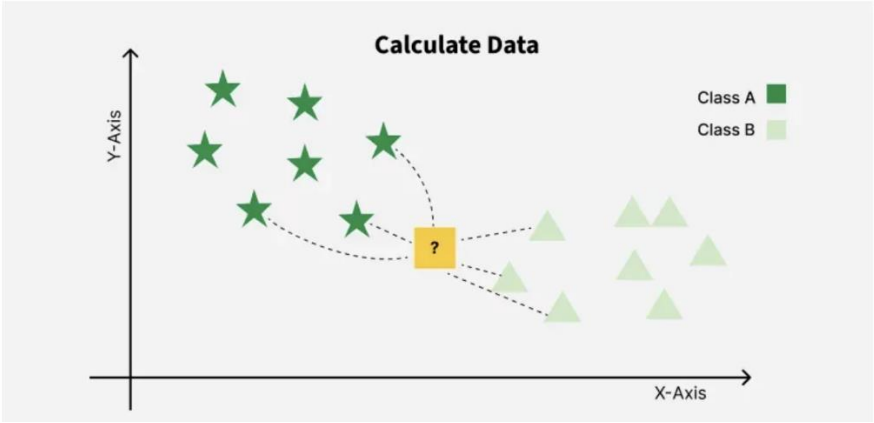
- **Supervised** → it learns from labeled data
- **Non-parametric** → it does not assume a fixed data distribution
- **Lazy learner** → it does not build a model during training; it stores the data and does work at prediction time

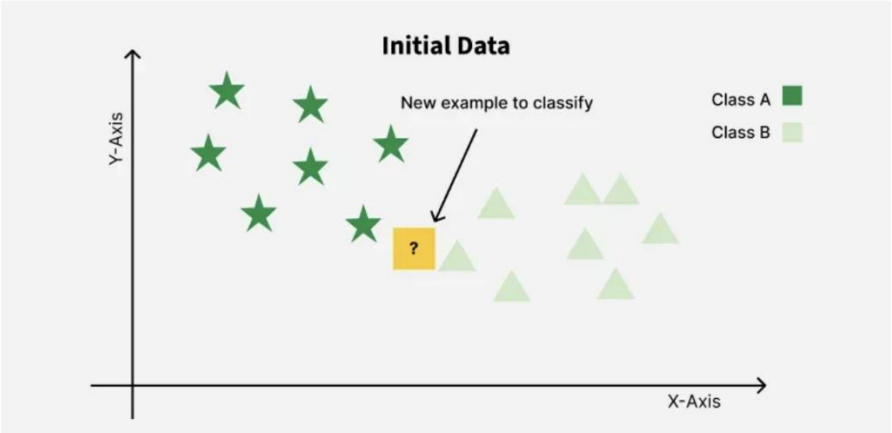
How KNN Works

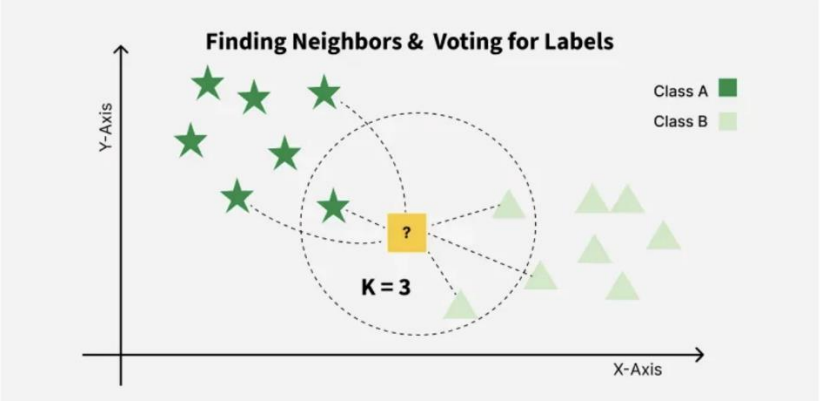
Step-by-step

1. Choose the value of **k**
2. Calculate the distance from the new point to all training points
3. Find the **k nearest neighbors**

4. For **classification**: choose the class with the majority vote
5. For **regression**: take the average of the neighbors' values







KNN for Classification

In classification, KNN predicts a **category**.

Example:

Suppose the 5 nearest neighbors of a new point are:

- 3 Red
- 2 Blue

Then the new point is classified as **Red** because Red is the majority.

KNN for Regression

In regression, KNN predicts a **continuous value**.

Example:

Suppose the 3 nearest neighbors have values:

- 20
- 25
- 30

Prediction:

$$\frac{20 + 25 + 30}{3} = 25$$

So the predicted value is **25**.

Distance Metrics in KNN

KNN needs a way to measure closeness.

1. Euclidean Distance

This is the **straight-line distance** between two points.
It is the most common distance metric in KNN.

Euclidean Distance

$$d(x$$

$$,y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

2. Manhattan Distance

This measures distance by moving only in horizontal and vertical directions, like city blocks.

Manhattan Distance

$$d(x, y) = \sum_{i=1}^{n} | x_i - y_i |$$

3. Minkowski Distance

This is a more general distance formula that includes Euclidean and Manhattan as special cases.

Minkowski Distance

$$d(x, y) = (\sum_{i=1}^{n} | x_i - y_i |^p)^{1/p}$$

4. Hamming Distance

Used mainly for strings or binary values, where distance is based on positions that differ.

Hamming Distance

$$d(x, y) = \sum_{i=1}^{n} \mathbf{1}(x_i \neq y_i)$$

Choosing the Value of K

The value of **k** is very important.

Small k

- more sensitive to noise
- can overfit
- high variance

Large k

- smoother predictions
- less affected by noise
- can underfit
- high bias

Common tip:

Use an **odd value** of k in classification to reduce ties.
Cross-validation is often used to find the best k.

Example

Imagine you want to identify whether a fruit is an **apple** or **banana** based on size and color.

A new fruit comes in.

If its 3 nearest neighbors are:

- apple
- apple
- banana

Then KNN predicts **apple**.

Why?
Because most nearby fruits are apples.

Advantages of KNN

- simple and easy to understand
- easy to implement

- works for both classification and regression
- no real training phase
- adapts easily when new data is added

Disadvantages of KNN

- slow for large datasets because it compares with many points
- uses a lot of memory
- sensitive to irrelevant features
- does not work well with high-dimensional data
- choice of k strongly affects performance
- may overfit when k is too small

Where KNN is Used

KNN is used in:

- recommendation systems
- pattern recognition
- spam detection
- healthcare prediction
- finance
- text and digit classification

Support Vector Machine (SVM)

What is SVM?

Support Vector Machine (SVM) is a **supervised machine learning algorithm** used mainly for **classification**, but it can also be used for **regression**.

Its main job is to find the **best boundary** that separates classes in the data. This boundary is called a **hyperplane**.

Main idea of SVM

SVM tries to draw a line or boundary between classes in such a way that the **distance between the boundary and the nearest points from both classes is as large as possible**.

This distance is called the **margin**.

Simple idea:

SVM finds the best separating boundary with the maximum margin.

Important Terms in SVM

1. Hyperplane

A **hyperplane** is the decision boundary that separates two classes.

- in 2D → it looks like a line
- in 3D → it looks like a plane
- in higher dimensions → it is called a hyperplane

2. Support Vectors

Support vectors are the data points closest to the hyperplane.

These points are very important because they help define the best boundary.
If support vectors move, the hyperplane also changes.

3. Margin

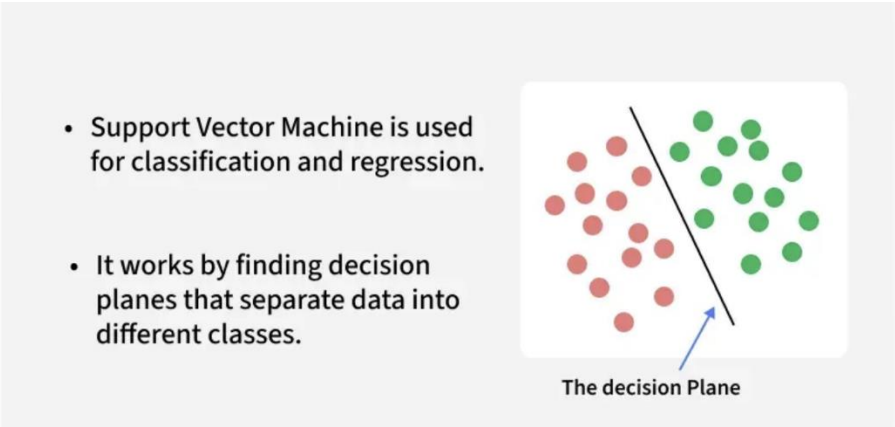
The **margin** is the distance between the hyperplane and the nearest data points from each class.

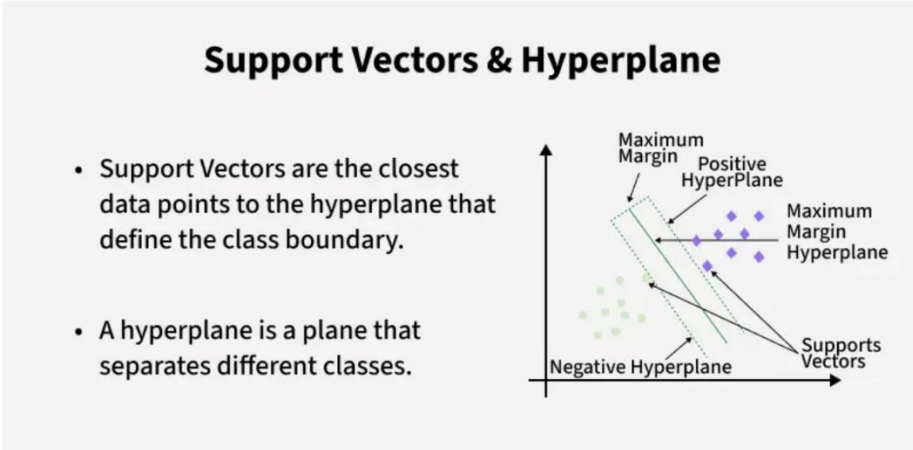
SVM tries to **maximize this margin** because a larger margin usually gives better generalization on unseen data.

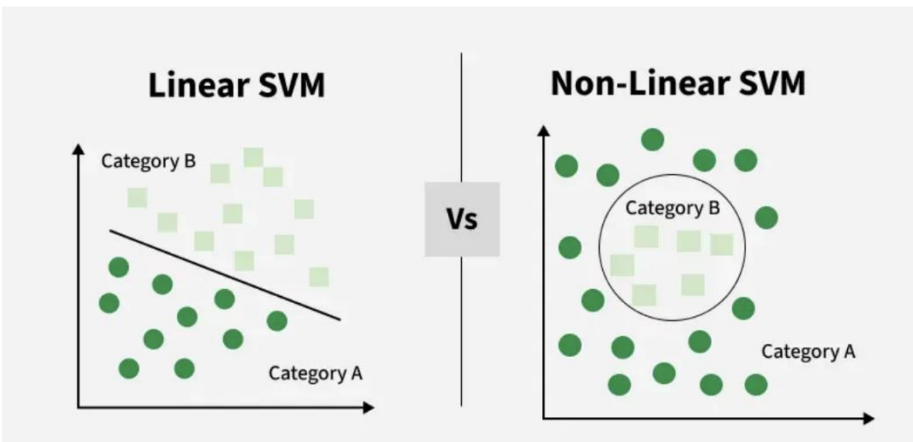
How SVM Works

Step-by-step

1. Take the labeled training data
2. Look for possible boundaries that can separate the classes
3. Find the boundary with the **largest margin**
4. Use the support vectors to define that boundary
5. Classify new data based on which side of the boundary it falls on







Types of SVM

1. Linear SVM

Used when the data can be separated by a straight line or flat boundary.

Example:

- spam vs not spam
- yes vs no
- pass vs fail

If the classes are clearly separable, linear SVM works well.

2. Non-Linear SVM

Used when data cannot be separated by a straight line.

In this case, SVM uses something called a **kernel** to transform the data into a higher-dimensional space where separation becomes easier.

Kernel

A **kernel** helps SVM handle non-linear data without explicitly doing complex transformations.

This idea is called the **kernel trick**.

Common kernels:

- Linear kernel
- Polynomial kernel
- RBF (Radial Basis Function) kernel
- Sigmoid kernel

Hard Margin vs Soft Margin

Hard Margin

Used when the data is perfectly separable.
SVM tries to separate all points with **no misclassification**.

Soft Margin

Used when the data has overlap, noise, or outliers.
SVM allows some misclassification in order to create a better and more practical boundary.

What is C in SVM?

C is a regularization parameter.

It controls the trade-off between:

- maximizing the margin
- minimizing classification errors

Simple rule:

- **Large C** → fewer mistakes allowed, narrower margin
- **Small C** → more mistakes allowed, wider margin

What is Gamma?

Gamma is important mainly in kernels like **RBF**.

It controls how much influence one training point has.

Simple idea:

- **High gamma** → focuses closely on nearby points
- **Low gamma** → gives smoother, wider influence

SVM for Regression

SVM can also be used for regression, called **Support Vector Regression (SVR)**.

Instead of separating classes, SVR tries to fit a line or function while keeping errors within an acceptable range.
It is used for continuous output values.

Example to Explain in Class

Suppose you want to classify emails as:

- Spam
- Not Spam

SVM looks for the best boundary that separates spam emails from non-spam emails.

It does not just separate them — it chooses the boundary with the **maximum margin**, so future emails can be classified better.

Why SVM is Powerful

SVM is strong because:

- it works well in high-dimensional data

- it focuses only on support vectors
- it can handle both linear and non-linear classification
- it often generalizes well to new data

Advantages of SVM

- works well for high-dimensional data
- effective for both linear and non-linear problems
- less prone to overfitting in many cases
- robust because it focuses on support vectors
- can be used for classification and regression

Disadvantages of SVM

- slow for very large datasets
- choosing the right kernel can be difficult
- sensitive to feature scaling
- harder to interpret than simpler models
- may struggle with noisy or overlapping data

Where SVM is Used

SVM is commonly used in:

- text classification
- sentiment analysis
- spam detection
- image classification
- bioinformatics
- medical diagnosis

- pattern recognition